

## Splunk SIEM Home Lab

### SOC Investigation with Boss of the SOC (BOTS) v3 Dataset

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## Objective

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I set up Splunk Enterprise on my Windows machine and loaded the Boss of the SOC (BOTS) v3 dataset, a free pre-built collection of realistic attack logs distributed by Splunk. The goal was to get hands-on experience using SPL (Search Processing Language) to investigate simulated attacks.

The lab covers installing Splunk, loading and verifying the dataset, running four investigation queries to identify brute force attempts, network scanning, web attacks, and suspicious process execution, building a detection dashboard, and mapping all findings to MITRE ATT&CK techniques.

## Tools Used

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- Splunk Enterprise: SIEM platform used to ingest, search, and visualize the attack data
- BOTS v3 Dataset: pre-indexed attack log dataset from Splunk containing Windows event logs, Sysmon, HTTP, and network traffic
- SPL (Search Processing Language): Splunk's query language used to search and analyze the log data
- Windows: host machine where Splunk was installed and run
- MITRE ATT&CK Framework: used to map detected activity to known adversary techniques

## Key Findings

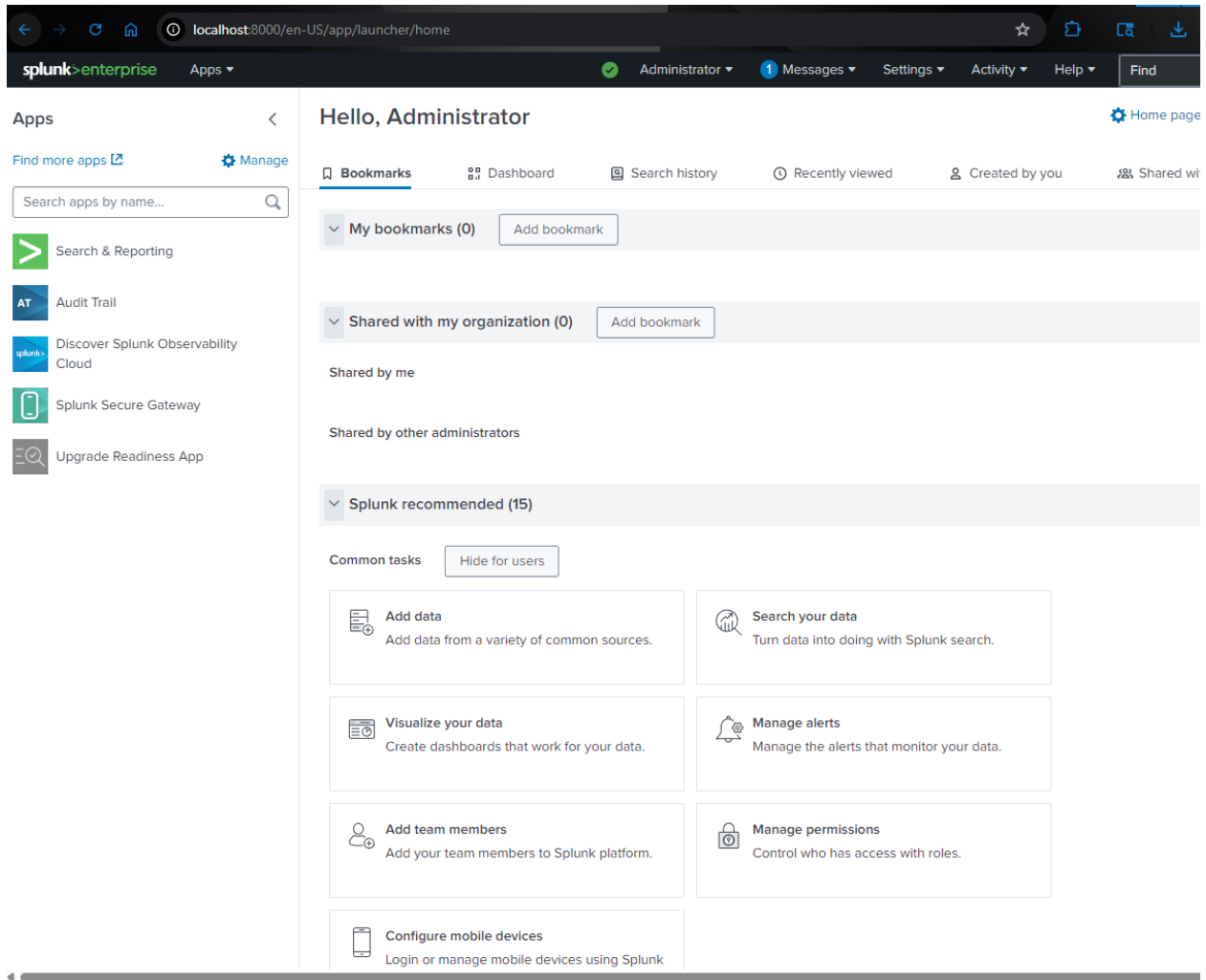
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- Identified brute force login activity by correlating EventCode 4625 (failed login) and 4624 (successful login) events, mapped to MITRE ATT&CK T1110
- Found a single source IP responsible for high-volume connection attempts across multiple destination IPs and ports, consistent with network scanning (T1046)
- Detected a source IP making an abnormally high number of POST requests to the same URI, indicating a web application attack mapped to T1190
- Identified suspicious process creation events via Sysmon EventCode 1 showing command-line activity consistent with attacker behavior, mapped to T1059
- Built a two-panel SOC dashboard in Splunk combining login anomaly and web attack visibility into a single view
- The BOTS v3 dataset loaded with approximately 107 sourcetypes, providing realistic multi-source log coverage across endpoint and network data

# 1. Installing Splunk Enterprise

I started by downloading and installing Splunk Enterprise on my Windows machine

1. Once installation finished, opened a browser and navigated to <http://localhost:8000>
2. Logged in with the credentials I set during installation and the Splunk home dashboard loaded



## 2. Downloading the BOTS v3 Dataset

The Boss of the SOC v3 dataset is a free, pre-indexed collection of attack logs made by Splunk. It simulates a realistic cyber attack across Windows endpoints, web servers, and network traffic. Because the data comes pre-indexed, it does not count against the Splunk trial license limits.

1. Went to [github.com/splunk/botsv3](https://github.com/splunk/botsv3)
2. Downloaded the `botsv3_data_set.tgz` file listed at the top of the page

3. Reviewed the Required Software section on the GitHub page and noted any add-ons listed

| Download                        |                 |         |                    |                                  |
|---------------------------------|-----------------|---------|--------------------|----------------------------------|
| Dataset                         | Description     | Size    | Format             | MD5                              |
| <a href="#">BOTS V3 Dataset</a> | BOTSv3 dataset. | 320.1MB | Pre-indexed Splunk | d7ccca99a01cff070dff3c139cdc10eb |

### 3. Loading the Dataset into Splunk

With the dataset downloaded, I couldn't load it directly into Splunk through the web interface so I inserted directly into the files.

1. In files I dragged the unzipped and extracted botsv3\_data\_set.tgz file into the path C:\Program Files\Splunk\etc\apps\botsv3\_data\_set
2. I then reset splunk process and index the data
3. Once complete, opened the Search and Reporting app and ran the following verification query:

**index=botsv3 earliest=0**

The query returned results and showed approximately 100+ sourcetypes, which confirmed the dataset loaded correctly. If it had returned nothing, the fix would have been to go to Settings > Server Controls > Restart Splunk and try the query again.

| splunk>enterprise Apps Administrator Messages Settings Activity Help Find                                                                                   |                             |        |                 |              |           |             |                |                |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------|-----------------|--------------|-----------|-------------|----------------|----------------|-----------|
| Indexes                                                                                                                                                     |                             |        |                 |              |           |             |                |                | New Index |
| A repository for data in Splunk Enterprise. Indexes reside in flat files on the Splunk Enterprise instance known as the indexer. <a href="#">Learn more</a> |                             |        |                 |              |           |             |                |                |           |
| Showing 11-17 of 17 Indexes                                                                                                                                 |                             |        |                 |              |           |             |                |                |           |
| Search...                                                                                                                                                   | 10 Per Page < Prev 1 2 Next |        |                 |              |           |             |                |                |           |
| Name                                                                                                                                                        | Status                      | Type   | App             | Current Size | Max Size  | Event Count | Earliest Event | Latest Event   |           |
| <a href="#">_telemetry</a>                                                                                                                                  | Active                      | Events | system          | 1 MB         | 488.28 GB | 8           | 2 hours ago    | 37 minutes ago |           |
| <a href="#">_thefishbucket</a>                                                                                                                              | Active                      | Events | system          | 1 MB         | 488.28 GB | 0           | —              | —              |           |
| <a href="#">botsv3</a>                                                                                                                                      | Active                      | Events | botsv3_data_set | 601 MB       | 488.28 GB | 2.03M       | 8 years ago    | 7 years ago    |           |
| <a href="#">history</a>                                                                                                                                     | Active                      | Events | system          | 1 MB         | 488.28 GB | 0           | —              | —              |           |
| <a href="#">main</a>                                                                                                                                        | Active                      | Events | system          | 1 MB         | 488.28 GB | 0           | —              | —              |           |
| <a href="#">splunklogger</a>                                                                                                                                | Inactive                    | Events | system          | 0 B          | 488.28 GB | 0           | —              | —              |           |
| <a href="#">summary</a>                                                                                                                                     | Active                      | Events | system          | 1 MB         | 488.28 GB | 0           | —              | —              |           |

**splunk>enterprise** Apps Admin... Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules Search & Report

## New Search

Save As Create Table View Close

index=botsv3 earliest=0 Time range: Last 24 hours

611,809 of 553,281 events matched No Event Sampling Job

Events (611,809) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 month per col

Format Show: 20 Per Page View: List

< Prev 1 2 3 4 5 6 7 8 ... Ne

| < Hide Fields      | All Fields | i | Time                       | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|------------|---|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SELECTED FIELDS    |            | > | 9/19/19<br>2:10:50.000 PM  | <error><message>Splunkd daemon is not responding: ('Error connecti<br>host = OD-FM-NA-i-0ad2d665d4bdace22.amazonaws.com   source = es                                                                                                                                                                                                                                                                                                                               |
| INTERESTING FIELDS |            | > | 8/20/18<br>11:27:09.296 AM | { [-]<br>bytes: 50637<br>bytes_in: 30<br>bytes_out: 50607<br>dest_ip: 172.16.0.178<br>dest_mac: 02:A0:38:1B:B3:70<br>endtime: 2018-08-20T15:27:09.296788Z<br>flow_id: c240fdab-997d-4285-a221-1080976a6b8a<br>fragment_count: 0<br>packets_in: 1<br>packets_out: 36<br>protocol: UDP<br>protocol_stack: ip:udp:unknown<br>protoid: 17<br>src_ip: 13.125.33.130<br>src_mac: 02:4F:D7:61:53:04<br>timestamp: 2018-08-20T15:27:09.296118Z<br>tos: 0<br>version: 4<br>} |

## 4. Running SPL Investigation Queries

This was the core of the lab. I ran four separate SPL queries to investigate different aspects of the attack captured in the dataset. For each one I reviewed the output, identified the suspicious activity, and mapped it to a MITRE ATT&CK technique.

### 4.1 Brute Force Login Detection

I started by looking for brute force login activity. EventCode 4625 is a failed login and EventCode 4624 is a successful login. Grouping them into transactions by host and account

name lets you spot an account that had a flood of failures followed by a success, which is the pattern of a successful brute force.

1. I pasted the following into the search bar

```
index=botsv3 sourcetype="WinEventLog:Security" (EventCode=4624 OR EventCode=4625)
| transaction host, AccountName maxspan=30m
| table _time AccountName host EventCode duration
```

2. I then set the time range to all time since the data set was made in 2018

The results showed login transactions grouped by host and user within 30-minute windows. I looked for accounts with a high count of 4625 events followed by a 4624, which would indicate a successful brute force. This activity maps to MITRE ATT&CK T1110 (Brute Force).

NEW SEARCH

index=botsv3 sourcetype="WinEventLog:Security" (EventCode=4624 OR EventCode=4625)  
| transaction host, AccountName maxspan=30m  
| table \_time AccountName host EventCode duration

Time range: All time

✓ 59 events (before 3/31/26 10:20:16.000 PM) No Event Sampling

Events Patterns **Statistics (59)** Visualization

Show: 20 Per Page Format Preview: On

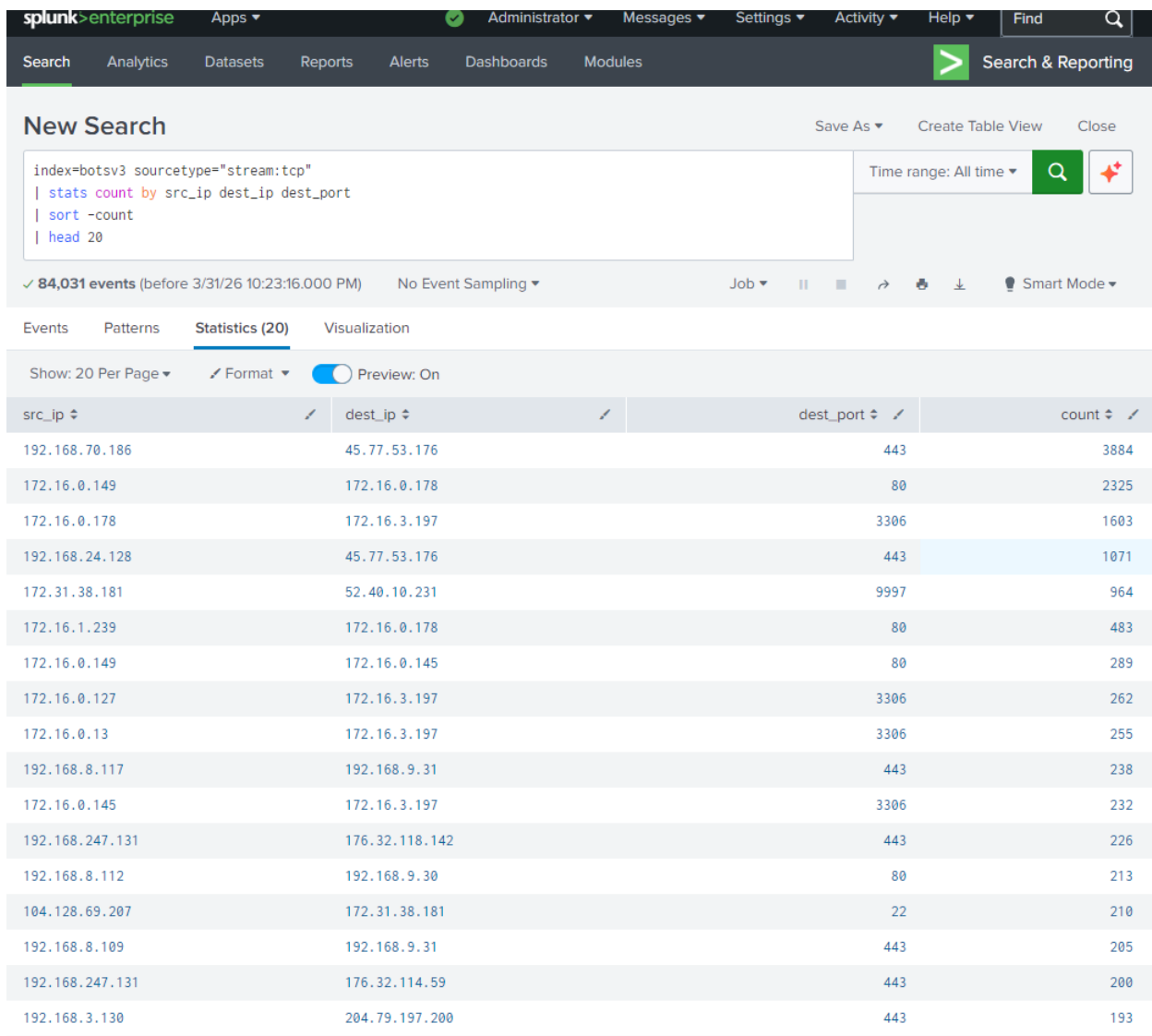
| _time               | AccountName | host      | EventCode    | duration |
|---------------------|-------------|-----------|--------------|----------|
| 2018-08-20 07:32:06 |             | MKRAEUS-L | 4624<br>4625 | 1495     |
| 2018-08-20 05:24:48 |             | SEPM      | 4624<br>4625 | 295      |
| 2018-08-20 10:56:11 |             | PCERF-L   | 4624         | 1243     |
| 2018-08-20 10:51:59 |             | JWORTOS-L | 4624         | 393      |
| 2018-08-20 10:51:16 |             | BTUN-L    | 4624         | 0        |
| 2018-08-20 10:51:11 |             | BSTOLL-L  | 4624         | 1526     |
| 2018-08-20 10:57:39 |             | FYODOR-L  | 4624         | 911      |
| 2018-08-20 10:55:23 |             | BGIST-L   | 4624         | 0        |
| 2018-08-20 10:05:58 |             | BTUN-L    | 4624         | 709      |
| 2018-08-20 10:02:48 |             | BSTOLL-L  | 4624         | 1450     |
| 2018-08-20 09:57:06 |             | PCERF-L   | 4624         | 1675     |
| 2018-08-20 09:57:01 |             | MKRAEUS-L | 4624         | 1552     |
| 2018-08-20 09:56:59 |             | JWORTOS-L | 4624         | 1676     |
| 2018-08-20 09:35:12 |             | BTUN-L    | 4624         | 341      |
| 2018-08-20 09:29:03 |             | BSTOLL-L  | 4624         | 1650     |
| 2018-08-20 09:28:47 |             | JWORTOS-L | 4624         | 1306     |
| 2018-08-20 09:23:04 |             | PCERF-L   | 4624         | 1639     |
| 2018-08-20 09:22:42 |             | MKRAEUS-L | 4624         | 1236     |
| 2018-08-20 09:11:51 |             | BGIST-L   | 4624         | 1199     |

## 4.2 Network Scanning Detection

Next I looked for signs of network scanning by finding source IPs that were connecting to an unusually large number of destinations. A normal host connects to a small set of IPs. A scanner touches many.

```
index=botsv3 sourcetype="stream:tcp"
| stats count by src_ip dest_ip dest_port
| sort -count
| head 20
```

The top results showed one source IP appearing across many different destination IPs and ports, which is the signature of an active network scanner. This maps to MITRE ATT&CK T1046 (Network Service Discovery).



The screenshot displays the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a secondary navigation bar with 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', 'Dashboards', and 'Modules'. A 'Search & Reporting' button is on the right.

The main content area is titled 'New Search'. It contains a search bar with the query: `index=botsv3 sourcetype="stream:tcp" | stats count by src_ip dest_ip dest_port | sort -count | head 20`. To the right of the search bar, there's a 'Time range: All time' dropdown and a search icon. Below the search bar, it shows '84,031 events (before 3/31/26 10:23:16.000 PM)' and 'No Event Sampling'.

The results are displayed in a table with the following columns: `src_ip`, `dest_ip`, `dest_port`, and `count`. The table is sorted by `count` in descending order. The first row shows `src_ip: 192.168.70.186`, `dest_ip: 45.77.53.176`, `dest_port: 443`, and `count: 3884`. The second row shows `src_ip: 172.16.0.149`, `dest_ip: 172.16.0.178`, `dest_port: 80`, and `count: 2325`. The third row shows `src_ip: 172.16.0.178`, `dest_ip: 172.16.3.197`, `dest_port: 3306`, and `count: 1603`. The fourth row shows `src_ip: 192.168.24.128`, `dest_ip: 45.77.53.176`, `dest_port: 443`, and `count: 1071`. The fifth row shows `src_ip: 172.31.38.181`, `dest_ip: 52.40.10.231`, `dest_port: 9997`, and `count: 964`. The sixth row shows `src_ip: 172.16.1.239`, `dest_ip: 172.16.0.178`, `dest_port: 80`, and `count: 483`. The seventh row shows `src_ip: 172.16.0.149`, `dest_ip: 172.16.0.145`, `dest_port: 80`, and `count: 289`. The eighth row shows `src_ip: 172.16.0.127`, `dest_ip: 172.16.3.197`, `dest_port: 3306`, and `count: 262`. The ninth row shows `src_ip: 172.16.0.13`, `dest_ip: 172.16.3.197`, `dest_port: 3306`, and `count: 255`. The tenth row shows `src_ip: 192.168.8.117`, `dest_ip: 192.168.9.31`, `dest_port: 443`, and `count: 238`. The eleventh row shows `src_ip: 172.16.0.145`, `dest_ip: 172.16.3.197`, `dest_port: 3306`, and `count: 232`. The twelfth row shows `src_ip: 192.168.247.131`, `dest_ip: 176.32.118.142`, `dest_port: 443`, and `count: 226`. The thirteenth row shows `src_ip: 192.168.8.112`, `dest_ip: 192.168.9.30`, `dest_port: 80`, and `count: 213`. The fourteenth row shows `src_ip: 104.128.69.207`, `dest_ip: 172.31.38.181`, `dest_port: 22`, and `count: 210`. The fifteenth row shows `src_ip: 192.168.8.109`, `dest_ip: 192.168.9.31`, `dest_port: 443`, and `count: 205`. The sixteenth row shows `src_ip: 192.168.247.131`, `dest_ip: 176.32.114.59`, `dest_port: 443`, and `count: 200`. The seventeenth row shows `src_ip: 192.168.3.130`, `dest_ip: 204.79.197.200`, `dest_port: 443`, and `count: 193`.

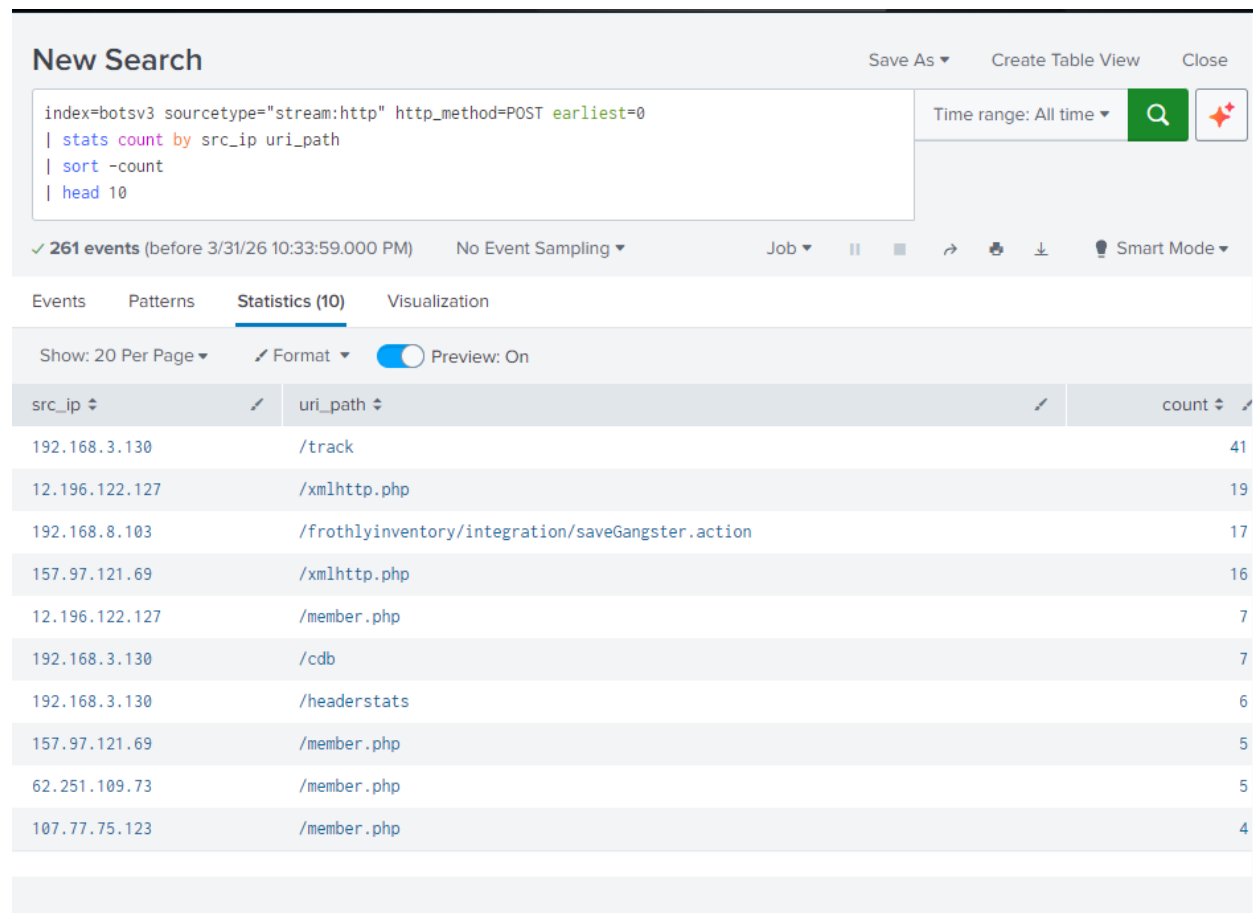
| src_ip          | dest_ip        | dest_port | count |
|-----------------|----------------|-----------|-------|
| 192.168.70.186  | 45.77.53.176   | 443       | 3884  |
| 172.16.0.149    | 172.16.0.178   | 80        | 2325  |
| 172.16.0.178    | 172.16.3.197   | 3306      | 1603  |
| 192.168.24.128  | 45.77.53.176   | 443       | 1071  |
| 172.31.38.181   | 52.40.10.231   | 9997      | 964   |
| 172.16.1.239    | 172.16.0.178   | 80        | 483   |
| 172.16.0.149    | 172.16.0.145   | 80        | 289   |
| 172.16.0.127    | 172.16.3.197   | 3306      | 262   |
| 172.16.0.13     | 172.16.3.197   | 3306      | 255   |
| 192.168.8.117   | 192.168.9.31   | 443       | 238   |
| 172.16.0.145    | 172.16.3.197   | 3306      | 232   |
| 192.168.247.131 | 176.32.118.142 | 443       | 226   |
| 192.168.8.112   | 192.168.9.30   | 80        | 213   |
| 104.128.69.207  | 172.31.38.181  | 22        | 210   |
| 192.168.8.109   | 192.168.9.31   | 443       | 205   |
| 192.168.247.131 | 176.32.114.59  | 443       | 200   |
| 192.168.3.130   | 204.79.197.200 | 443       | 193   |

## 4.3 Web Attack Detection

I then searched for web application attack behavior by looking at POST requests. A normal user does not send hundreds of POST requests to the same endpoint in a short window. A high volume of POSTs to a single URI is consistent with SQL injection or credential stuffing.

```
index=botsv3 sourcetype="stream:http" http_method=POST earliest=0
| stats count by src_ip uri_path
| sort -count
| head 10
```

One source IP stood out immediately with a very high count of POST requests all targeting the same URI. This maps to MITRE ATT&CK T1190 (Exploit Public-Facing Application).



The screenshot shows the Splunk Search interface. At the top, the search bar contains the query: `index=botsv3 sourcetype="stream:http" http_method=POST earliest=0 | stats count by src_ip uri_path | sort -count | head 10`. The results are displayed in a table with columns `src_ip`, `uri_path`, and `count`. The first result shows 41 events for `src_ip: 192.168.3.130` and `uri_path: /track`. Other results include `/xmlhttp.php` (19 events), `/frothlyinventory/integration/saveGangster.action` (17 events), and several other endpoints with counts ranging from 7 to 5.

| src_ip         | uri_path                                          | count |
|----------------|---------------------------------------------------|-------|
| 192.168.3.130  | /track                                            | 41    |
| 12.196.122.127 | /xmlhttp.php                                      | 19    |
| 192.168.8.103  | /frothlyinventory/integration/saveGangster.action | 17    |
| 157.97.121.69  | /xmlhttp.php                                      | 16    |
| 12.196.122.127 | /member.php                                       | 7     |
| 192.168.3.130  | /cdb                                              | 7     |
| 192.168.3.130  | /headerstats                                      | 6     |
| 157.97.121.69  | /member.php                                       | 5     |
| 62.251.109.73  | /member.php                                       | 5     |
| 107.77.75.123  | /member.php                                       | 4     |

## 4.4 Suspicious Process Execution

Finally I used Sysmon data to look at process creation events. Sysmon EventCode 1 logs every process that gets launched, including the full command line and the parent process that spawned it. This is one of the most useful event types for spotting attacker activity on an endpoint.

```
index=botsv3 sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
```

```
| table _time Computer User CommandLine ParentCommandLine  
| head 20
```

I reviewed the CommandLine and ParentCommandLine columns looking for things like cmd.exe or powershell.exe launching from a temp directory, which is a common attacker technique. The suspicious entries I found map to MITRE ATT&CK T1059 (Command and Scripting Interpreter).

The screenshot shows the Splunk Search & Reporting interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a secondary navigation bar with 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', 'Dashboards', and 'Modules'. The main content area is titled 'New Search' and contains a search bar with the following query: `index=botsv3 sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"`. Below the search bar, the query is formatted as `| table _time Computer User CommandLine ParentCommandLine` followed by `| head 20`. To the right of the search bar, there's a 'Time range: All time' dropdown and a search button. Below the search bar, it shows '9,212 events (before 3/31/26 10:29:15.000 PM)' and 'No Event Sampling'. The results are displayed in a table with columns: '\_time', 'Computer', 'User', 'CommandLine', and 'ParentCommandLine'. The table shows 20 rows of data, all with the same timestamp '2018-08-20 10:52:01'.

| _time               | Computer | User | CommandLine | ParentCommandLine |
|---------------------|----------|------|-------------|-------------------|
| 2018-08-20 10:52:02 |          |      |             |                   |
| 2018-08-20 10:52:02 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |
| 2018-08-20 10:52:01 |          |      |             |                   |

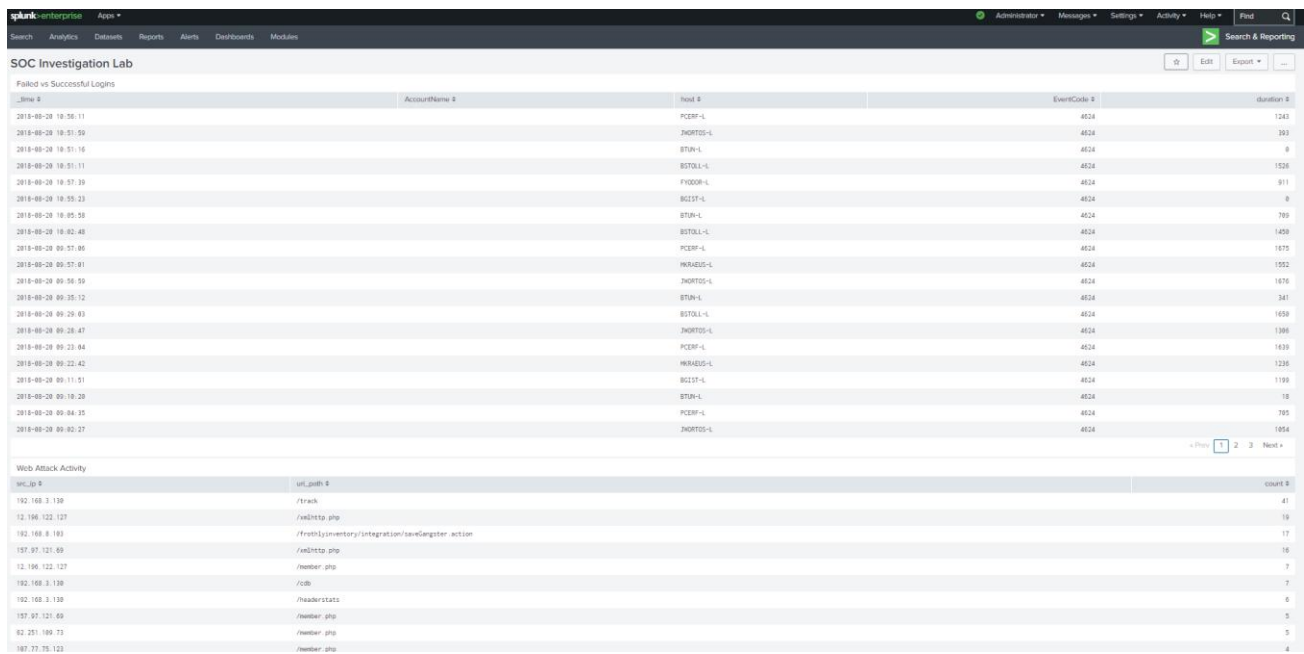
## 5. Building the SOC Dashboard

After running the queries, I saved two of them as dashboard panels to create a combined view that would let me monitor both login anomalies and web attack activity at the same time. This is closer to how a real SOC analyst would use Splunk during a shift.

1. Re-ran Query 1 (brute force detection) in Search and Reporting



- Clicked Save as
- Selected Dashboard Panel from the dropdown
- Created a new dashboard and named it SOC Investigation Lab
- Named the first panel Failed vs Successful Logins and saved it
- Went back to Search and Reporting and re-ran Query 3 (web attack detection)
- Clicked Save As > Dashboard Panel again and selected the existing SOC Investigation Lab dashboard
- Named the second panel Web Attack Activity and saved it
- Opened the completed dashboard and confirmed both panels were displaying data



The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk-enterprise' logo, 'Apps', and various menu items like 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a search bar and a 'Search & Reporting' button. The main content area displays a dashboard titled 'SOC Investigation Lab'. It contains two panels. The first panel, 'Failed vs Successful Logins', shows a table with columns: '\_time', 'AccountName', 'Host', 'EventCode', and 'duration'. It lists various login attempts with timestamps, usernames, hostnames, and event codes. The second panel, 'Web Attack Activity', shows a table with columns: 'src\_ip', 'url\_path', and 'count'. It lists various web requests with source IP addresses, URL paths, and counts.

| _time               | AccountName | Host      | EventCode | duration |
|---------------------|-------------|-----------|-----------|----------|
| 2018-08-28 18:56:11 |             | PCERF-L   | 4024      | 1243     |
| 2018-08-28 18:51:59 |             | ZKORTOS-L | 4024      | 163      |
| 2018-08-28 18:51:16 |             | STAN-L    | 4024      | 9        |
| 2018-08-28 18:51:11 |             | BITOLL-L  | 4024      | 1528     |
| 2018-08-28 18:57:39 |             | FIDOCOR-L | 4024      | 911      |
| 2018-08-28 18:55:23 |             | BC117-L   | 4024      | 9        |
| 2018-08-28 18:45:58 |             | STAN-L    | 4024      | 789      |
| 2018-08-28 18:40:48 |             | BITOLL-L  | 4024      | 1459     |
| 2018-08-28 09:57:06 |             | PCERF-L   | 4024      | 1675     |
| 2018-08-28 09:57:01 |             | MRAGSD-L  | 4024      | 1952     |
| 2018-08-28 09:56:59 |             | ZKORTOS-L | 4024      | 1676     |
| 2018-08-28 09:25:12 |             | STAN-L    | 4024      | 341      |
| 2018-08-28 09:29:03 |             | BITOLL-L  | 4024      | 1659     |
| 2018-08-28 09:28:47 |             | ZKORTOS-L | 4024      | 1390     |
| 2018-08-28 09:23:04 |             | PCERF-L   | 4024      | 1639     |
| 2018-08-28 09:22:42 |             | MRAGSD-L  | 4024      | 1228     |
| 2018-08-28 09:11:51 |             | BC117-L   | 4024      | 1199     |
| 2018-08-28 09:19:28 |             | STAN-L    | 4024      | 18       |
| 2018-08-28 09:04:35 |             | PCERF-L   | 4024      | 795      |
| 2018-08-28 09:02:27 |             | ZKORTOS-L | 4024      | 1854     |

| src_ip         | url_path                                        | count |
|----------------|-------------------------------------------------|-------|
| 192.168.3.139  | /track                                          | 41    |
| 12.196.122.127 | /redhttp.php                                    | 19    |
| 192.168.8.193  | /freshInventory/integration/saveLangster.action | 17    |
| 197.97.121.69  | /redhttp.php                                    | 16    |
| 12.196.122.127 | /member.php                                     | 7     |
| 192.168.3.139  | /cds                                            | 7     |
| 192.168.3.139  | /headerStats                                    | 6     |
| 197.97.121.69  | /member.php                                     | 5     |
| 62.251.190.73  | /member.php                                     | 5     |
| 197.72.75.123  | /member.php                                     | 4     |

## 6. MITRE ATT&CK Mapping

After completing all four queries, I looked up each detected technique on [attack.mitre.org](https://attack.mitre.org) and recorded the technique ID, name, and a brief description of what the activity represented.

| Activity Detected          | MITRE ID     | Description                                                                                                           |
|----------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|
| Brute Force Login Attempts | <b>T1110</b> | Adversaries use repeated authentication attempts to guess or crack credentials against target accounts.               |
| Network Scanning           | <b>T1046</b> | Adversaries enumerate network services and open ports to map the target environment before launching further attacks. |
| Web Application Attack     | <b>T1190</b> | Adversaries exploit vulnerabilities in public-facing web applications to gain initial access or extract data.         |

|                              |              |                                                                                                                    |
|------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------|
| Suspicious Process Execution | <b>T1059</b> | Adversaries use command-line interpreters like cmd.exe or PowerShell to execute malicious commands on an endpoint. |
|------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------|

## 7. Summary

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This lab covered a full Splunk SIEM investigation from installation through active threat analysis. I loaded a realistic pre-built attack dataset, used SPL to identify four distinct attack behaviors across endpoint and network data, and built a dashboard that surfaces the two most SOC-relevant findings into a single view.

The most useful part of the lab was working through the Sysmon process creation data. Seeing how a suspicious process chain looks in raw log form versus how it gets surfaced in a query output gives a clearer picture of what SOC analysts are actually doing when they say they are triaging alerts.

- Installed Splunk Enterprise and loaded the BOTS v3 dataset containing approximately 100+ sourcetypes
- Ran four SPL queries across Windows Security logs, Sysmon, HTTP stream, and TCP stream data
- Identified brute force activity (T1110), network scanning (T1046), web exploitation (T1190), and suspicious process execution (T1059)
- Built a two-panel SOC Investigation Lab dashboard combining login and web attack visibility
- Mapped all findings to MITRE ATT&CK and documented indicators of compromise from the dataset